

Chapter 02: Polynomial Interpolation

Polynomials:

General Form:

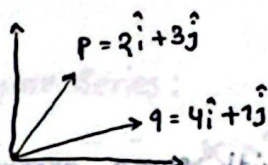
$$P_n(x) = \underbrace{a_0}_{\text{Degree}} x^0 + \underbrace{a_1}_{\text{coeff}} x^1 + \underbrace{a_2}_{\text{coeff}} x^2 + \dots + \underbrace{a_n}_{\text{coeff}} x^n$$

$$P_3(x) = a_0 x^0 + a_1 x^1 + a_2 x^2 + a_3 x^3$$

coeff = $[a_0, a_1, a_2, a_3] \Rightarrow 4 = n+1$

$$\text{Number of coefficient} = \text{Degree} + 1$$

Vector Space: A region where we can add or multiply with scalars.



$$p + q = 6\hat{i} + 4\hat{j}$$

$$2 * p = 4\hat{i} + 6\hat{j}$$

$$p + (-q) = -2\hat{i} + 2\hat{j}$$

Polynomial:

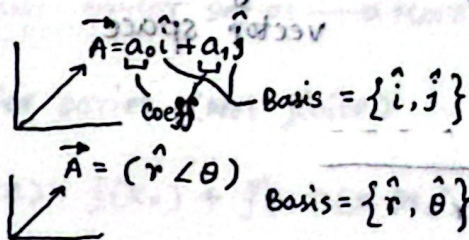
$$P_2(x) = 1 + x + 2x^2$$

$$P_2(x) + P_3(x) = 1 + x + 2x^2 + x^3$$

$$P_3(x) = x^3$$

$$P_3(x) * 5 = 5x^3$$

Basis: A set of vectors that spans the space.



Polynomial:

$$P_3(x) = a_0 x^0 + a_1 x^1 + a_2 x^2 + a_3 x^3$$

coeff

$$\text{Basis} = \{x^0, x^1, x^2, x^3\} = \{1, x, x^2, x^3\}$$

Natural Basis

⇒ Dimensional Space: How many dimensions?

For 3 degree Polynomials; Basis = $\{1, x, x^2, x^3\}$

Basis Dimension = 4 $x_0 + x_1 + x_2 + x_3 = (x)$

* **Basis Dimension = Degree + 1**

Ex: $P_{37}(x) = \dots$; Degree, Number of coeff, basis, Dimension?
 $\downarrow \quad \downarrow \quad \downarrow \quad \downarrow$
 $n=37 \quad n+1=38 \quad \{1, x, x^2, \dots, x^{37}\} \quad n+1=38$

⇒ Functional Space:

→ Natural function $\rightarrow$ has ∞ degree, ∞ dimension

let, $f(x) = 2 + 3x + 10x^2 + 14x^3 + \dots$

basis = $\{1, x, x^2, x^3, \dots\}$

We can reproduce it using polynomials with some errors.

$P_2(x) = 2 + 3x + 10x^2 \quad | \quad f(x) = P_2(x) + \text{Error}$

$f(x) \in V^\infty \leftarrow$ function belongs to ∞ dimension vector space

$P_n(x) \in V^{n+1} \leftarrow$ Polynomial with degree n , belongs to $(n+1)$ dimension vector space

$f(x) = \underbrace{2 + 3x + 10x^2}_{P_2(x)} + \underbrace{14x^3 + \dots}_{\text{Error}}$

Error $\rightarrow$ Truncation Error

$P_2(x) = a_0 + a_1x + a_2x^2$

$P_5(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + a_5x^5$

* Higher the degree is, lower the error becomes.

$n \uparrow \rightarrow \text{error} \downarrow$

This is called,

Weierstrass Approximation Theorem:

$$\text{Error} = |f(x) - P_n(x)| \leq \epsilon$$

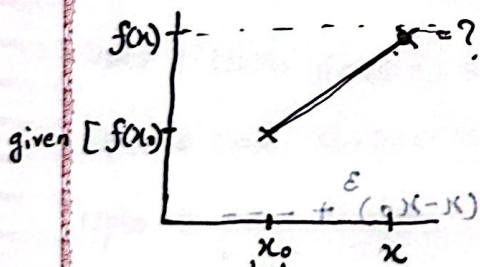
For any $f \in C([0,1])$ and any $\epsilon > 0$, there exist a polynomial

$P(x)$ such that

$$\max_{0 \leq x \leq 1} |f(x) - P_n(x)| < \epsilon$$

$$f(x) \approx P_n(x)$$

⇒ Taylor Series:



$$f'(x_0) = \frac{f(x_2) - f(x_1)}{x_2 - x_1}$$

$$\Rightarrow f'(x_0) = \frac{f(x) - f(x_0)}{x - x_0}$$

$$f(x) = f(x_0) + f'(x_0)(x - x_0)$$

Taylor series for straight line:

Actual Taylor series → More complex & infinitely large.

Taylor series, (Not finite)

$$f(x) = f(x_0) + f'(x_0)(x - x_0) + \frac{1}{2!} f''(x_0)(x - x_0)^2 + \frac{1}{3!} f'''(x_0)(x - x_0)^3 + \dots$$

Q. Proof of Taylor series: the higher the degree is, lower the error

Step 1:

$$f(x) = a_0 + a_1(x-x_0) + a_2(x-x_0)^2 + a_3(x-x_0)^3 + \dots$$

$$f'(x) = 0 + a_1 + 2a_2(x-x_0) + 3a_3(x-x_0)^2 + \dots$$

$$f''(x) = 0 + 0 + 2a_2 + 3 \times 2a_3(x-x_0) + \dots$$

$$f'''(x) = 0 + 0 + 0 + 3 \times 2a_3 + \dots$$

Step 2:

let, $x = x_0$

$$f(x_0) = a_0 + a_1(x_0-x_0) + a_2(x_0-x_0)^2 + a_3(x_0-x_0)^3 + \dots$$

$$f(x_0) = a_0$$

$$\text{Now, } a_0 = f(x_0)$$

$$f'(x_0) = a_1$$

$$a_1 = f'(x_0)$$

$$f''(x_0) = 2a_2$$

$$a_2 = \frac{f''(x_0)}{2!}$$

$$f'''(x_0) = 3 \times 2a_3 \Rightarrow a_3 = \frac{f'''(x_0)}{3!}$$

$$f(x) = f(x_0) + f'(x_0)(x-x_0) + \frac{1}{2!} f''(x_0)(x-x_0)^2 + \frac{1}{3!} f'''(x_0)(x-x_0)^3 + \dots$$

$$\therefore f(x) = f(x_0) + f'(x_0)(x-x_0) + \frac{1}{2!} f''(x_0)(x-x_0)^2 + \frac{1}{3!} f'''(x_0)(x-x_0)^3 + \dots$$

(Proved)

(not finite)

$$f(x) = f(x_0) + f'(x_0)(x-x_0) + \frac{1}{2!} f''(x_0)(x-x_0)^2 + \frac{1}{3!} f'''(x_0)(x-x_0)^3 + \dots$$

Example: Taylor series expansion of $\sin(x)$

$$f(x) = f(x_0) + f'(x_0)(x-x_0) + \frac{1}{2!} f''(x_0)(x-x_0)^2 + \dots$$

let $x_0 = 0$

$$f(x) = \sin(0) + \cos(0)(x) + \frac{1}{2!} (-\sin(0))(x^2) + \frac{1}{3!} (-\cos(0))(x^3) + \frac{1}{4!} \sin(0)x^4 + \frac{1}{5!} \cos(0)x^5 + \dots$$

$$= 0 + x + 0 - \frac{x^3}{3!} + 0 + \frac{x^5}{5!} + \dots$$

$$\therefore \sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots$$

Example: Find $\sin(0.1)$ using Taylor series

$$\sin x = \underbrace{x}_{1^{\text{st}} \text{ term}} - \underbrace{\frac{x^3}{3!}}_{2^{\text{nd}} \text{ term}} + \underbrace{\frac{x^5}{5!}}_{3^{\text{rd}} \text{ term}} - \underbrace{\frac{x^7}{7!}}_{4^{\text{th}} \text{ term}} + \dots$$

Upto 1 term, $\sin(0.1) \approx 0.1$

Upto 2 term, $\sin(0.1) \approx 0.1 - \frac{(0.1)^3}{3!} = 0.0998333$

Upto 3 term, $\sin(0.1) \approx 0.1 - \frac{(0.1)^3}{3!} + \frac{(0.1)^5}{5!} = 0.09983341667 \dots$

Actual value: $\sin(0.1) = 0.09983341665 + \dots$

$$\begin{bmatrix} 0.1 \\ 0.02 \\ 0.002 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

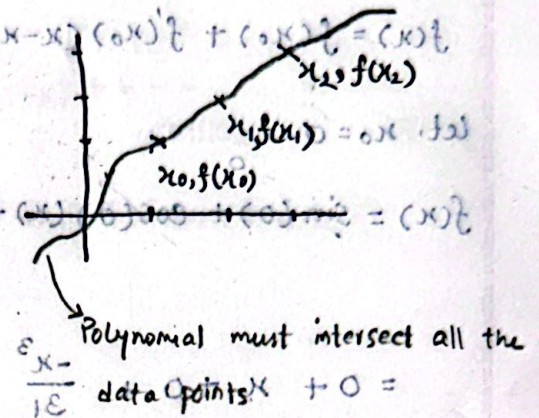
$$Y^T X = A \Leftrightarrow Y = A \cdot X$$

Polynomial Interpolation:

Given $(n+1)$ datapoints

house size	Rent
$x_0 = 20$	$f(x_0) = 200$
$x_1 = 30$	$f(x_1) = 250$
$x_2 = 40$	$f(x_2) = 270$

$$P_n(x) = P_2(x) = a_0 x^0 + a_1 x^1 + a_2 x^2$$



* Vandermonde Matrix

Given,

time	velocity
$x_0 = 1$	$f(x_0) = y_0 = 40$
$x_1 = 2$	$f(x_1) = y_1 = 50$
$x_2 = 3$	$f(x_2) = y_2 = 55$

$$P_n(x) = P_2(x) = a_0 x^0 + a_1 x^1 + a_2 x^2$$

$$a_0 \cdot 1^0 + a_1 \cdot 1^1 + a_2 \cdot 1^2 = 40 \quad \text{--- (i)}$$

$$a_0 \cdot 2^0 + a_1 \cdot 2^1 + a_2 \cdot 2^2 = 50 \quad \text{--- (ii)}$$

$$a_0 \cdot 3^0 + a_1 \cdot 3^1 + a_2 \cdot 3^2 = 55 \quad \text{--- (iii)}$$

$$\begin{bmatrix} 1^0 & 1^1 & 1^2 \\ 2^0 & 2^1 & 2^2 \\ 3^0 & 3^1 & 3^2 \end{bmatrix} \cdot \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} 40 \\ 50 \\ 55 \end{bmatrix}$$

$X \quad A \quad Y$

$$\Rightarrow X \cdot A = Y \quad \Rightarrow A = X^{-1} Y$$

Example:

	Time	Velocity
Node = n+1 = 2	15	362.8
	20	517.3

degree = 1

$$P_1(x) = a_0 x^0 + a_1 x^1$$

$$P_1(15) = a_0 + a_1 \cdot 15 = 362.8$$

$$P_1(20) = a_0 + a_1 \cdot 20 = 517.3$$

$$(x-x_1)(x-x_2) + (x-x_2)(x-x_3) + (x-x_3)(x-x_4) + (x-x_4)(x-x_5) = (x-x_1)(x-x_2)$$

$$P_1(x) = -100.85 + 30.9x$$

$$XA = Y$$

$$\Rightarrow \begin{bmatrix} x_0^0 & x_0^1 \\ x_1^0 & x_1^1 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \end{bmatrix} = \begin{bmatrix} y_0 \\ y_1 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 1 & 15 \\ 1 & 20 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \end{bmatrix} = \begin{bmatrix} 362.8 \\ 517.3 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} a_0 \\ a_1 \end{bmatrix} = \begin{bmatrix} 1 & 15 \\ 1 & 20 \end{bmatrix}^{-1} \begin{bmatrix} 362.8 \\ 517.3 \end{bmatrix}$$

$$= \begin{bmatrix} 4 & -3 \\ -0.2 & 0.2 \end{bmatrix} \begin{bmatrix} 362.8 \\ 517.3 \end{bmatrix}$$

$$\begin{bmatrix} a_0 \\ a_1 \end{bmatrix} = \begin{bmatrix} -100.85 \\ 30.9 \end{bmatrix}$$

Example: Given the time and velocity $v(t)$, find an interpolating Polynomial of velocity that goes through the datapoints using Vandermonde Matrix. Also, find the approx value of acceleration at Time $t = 7$ second.

	Time (t)	Velocity, $v(t)$
x_0	3	11
x_1	5	21
x_2	7	26

$$\Rightarrow \text{Nodes} = n+1 = 3$$

$$\text{Degree} = 2$$

$$P_2(x) = a_0 x^0 + a_1 x^1 + a_2 x^2$$

$$XA = Y$$

$$\Rightarrow A = X^{-1}Y$$

$$\Rightarrow \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} 3^0 & 3^1 & 3^2 \\ 5^0 & 5^1 & 5^2 \\ 7^0 & 7^1 & 7^2 \end{bmatrix}^{-1} \begin{bmatrix} 11 \\ 21 \\ 26 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} -13.37 \\ 10 \\ -0.625 \end{bmatrix}$$

$$P_2(x) = -13.37 + 10x - 0.625x^2$$

$$\text{Acceleration} = \frac{d}{dx} P_2(x)$$

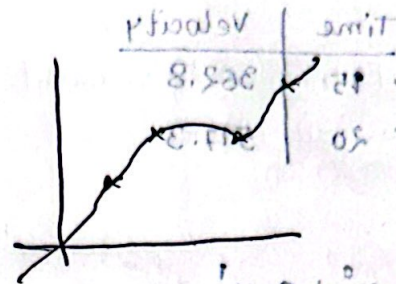
$$A(x) = 10 - 1.25x$$

$$\therefore A(7) = 1.25 \text{ ms}^{-2}$$

⇒ Lagrange Interpolation:

Example:

$Y = AX$
 $\begin{bmatrix} y_0 \\ y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} f(x_0) \\ f(x_1) \\ f(x_2) \\ f(x_3) \end{bmatrix}$
 $\text{Node } n+1 = 4$
 $\text{Degree} = 3$
 $P_3(x) = a_0x^0 + a_1x^1 + a_2x^2 + a_3x^3$



$$P(x) = a_0x^0 + a_1x^1 + a_2x^2 + a_3x^3$$

$$P(12) = a_0 + a_1 \cdot 12 + a_2 \cdot 12^2 + a_3 \cdot 12^3$$

$$P(10) = a_0 + a_1 \cdot 10 + a_2 \cdot 10^2 + a_3 \cdot 10^3$$

For Lagrange,

$$P_3(x) = L_0(x)f(x_0) + L_1(x)f(x_1) + L_2(x)f(x_2) + L_3(x)f(x_3)$$

Lagrange Basis

$$L_0(x) = \frac{x-x_1}{x_0-x_1} \cdot \frac{x-x_2}{x_0-x_2} \cdot \frac{x-x_3}{x_0-x_3}$$

→ Given

$$L_1(x) = \frac{x-x_0}{x_1-x_0} \cdot \frac{x-x_2}{x_1-x_2} \cdot \frac{x-x_3}{x_1-x_3}$$

$$L_2(x) = \frac{x-x_0}{x_2-x_0} \cdot \frac{x-x_1}{x_2-x_1} \cdot \frac{x-x_3}{x_2-x_3}$$

$$L_3(x) = \frac{x-x_0}{x_3-x_0} \cdot \frac{x-x_1}{x_3-x_1} \cdot \frac{x-x_2}{x_3-x_2}$$

$$\begin{bmatrix} -13.87 \\ 0 \\ 10 \\ -0.652 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}$$

$$P(x) = -13.87 + 10x - 0.652x^3$$

$$a(x) = \frac{b}{x^3}$$

$$a(x) = 10 - 1.32x$$

$$a(x) = 1.32x$$

$$N = 1 + N = 2060 \Rightarrow$$

$$\text{Degree} = 3$$

$$P(x) = a_0x^0 + a_1x^1 + a_2x^2 + a_3x^3$$

$$Y = AX$$

$$Y^T X = A^T$$

$$\begin{bmatrix} 1 \\ 12 \\ 10 \\ 12 \end{bmatrix} \begin{bmatrix} 1 \\ 12 \\ 10 \\ 12 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}$$

Example:

Time(x)	Velocity f(x)
x ₀ 15	227.04 f(x ₀)
x ₁ 20	362.78 f(x ₁)
x ₂ 22.5	517.35 f(x ₂)

What is the Lagrange Polynomial?
If x=17, Polynomial=?

$$\Rightarrow \text{Node} = n+1 = 3$$

$$\text{Degree} = 2$$

$$P_2(x) = L_0(x) \cdot f(x_0) + L_1(x) \cdot f(x_1) + L_2(x) \cdot f(x_2)$$

$$L_0(x) = \frac{x-x_1}{x_0-x_1} \cdot \frac{x-x_2}{x_0-x_2} = \frac{x-20}{15-20} \cdot \frac{x-22.5}{15-22.5} = \frac{2(x-20)(x-22.5)}{75}$$

$$L_1(x) = \frac{x-x_0}{x_1-x_0} \cdot \frac{x-x_2}{x_1-x_2} = \frac{x-15}{20-15} \cdot \frac{x-22.5}{20-22.5} = \frac{-2(x-15)(x-22.5)}{25}$$

$$L_2(x) = \frac{x-x_0}{x_2-x_0} \cdot \frac{x-x_1}{x_2-x_1} = \frac{x-15}{22.5-15} \cdot \frac{x-20}{22.5-20} = \frac{4(x-15)(x-20)}{75}$$

$$P_2(x) = \frac{2}{75}(x-20)(x-22.5)(227.04) - \frac{2}{25}(x-15)(x-22.5)(362.78) + \frac{4}{75}(x-15)(x-20)(517.35)$$

$$P_2(17) = \frac{2}{75}(17-20)(17-22.5)(227.04) - \frac{2}{25}(17-15)(17-22.5)(362.78) + \frac{4}{75}(17-15)(17-20)(517.35)$$

$$\therefore P_2(17) = 253.592$$

Example:

x	$f(x)$
0	0
10	227.4
15	362.8
20	517.35
22.5	602.97
30	90.67

Polynomial = ?
 $x=19$, $P_n(x)=?$; Allowed to use a polynomial of degree 2.

$$E = 1 + n = 6$$

$$\Rightarrow \text{Node} = n+1 = 6$$

As we will use $x=19$ and Allowed degree is 2.

$$\text{No Degree} = 5$$

So we will need to use 2+1=3 closest datapoint

$$\frac{(x-x_1)(x-x_2)}{(x_0-x_1)(x_0-x_2)} = \frac{x-20}{20-15} \cdot \frac{x-22.5}{15-22.5} = \frac{x-20}{5} \cdot \frac{x-22.5}{-7.5} = (x)_0$$

$$P_2(x) = l_0(x) \cdot f(x_0) + l_1(x) \cdot f(x_1) + l_2(x) \cdot f(x_2)$$

$$l_0(x) = \frac{x-x_1}{x_0-x_1} \cdot \frac{x-x_2}{x_0-x_2} = \frac{x-20}{15-20} \cdot \frac{x-22.5}{15-22.5} = \frac{2(x-20)(x-22.5)}{75}$$

$$l_1(x) = \frac{x-x_0}{x_1-x_0} \cdot \frac{x-x_2}{x_1-x_2} = \frac{x-0}{20-0} \cdot \frac{x-22.5}{20-22.5} = \frac{-2(x-15)(x-22.5)}{25}$$

$$l_2(x) = \frac{x-x_0}{x_2-x_0} \cdot \frac{x-x_1}{x_2-x_1} = \frac{x-0}{22.5-0} \cdot \frac{x-20}{22.5-20} = \frac{4(x-15)(x-20)}{75}$$

$$P_2(x) = \frac{2}{75}(x-20)(x-22.5)(362.8) - \frac{2}{25}(x-15)(x-22.5)(517.35) + \frac{4}{75}(x-15)(x-20)(602.97)$$

$$P_2(19) = 484.66$$

* Advantage

- No need to Inverse a matrix

* Disadvantage

- If new node is added or any node is removed the every parts need to be recalculated.

Newton's Divided Difference Form

x	$f(x)$
x_0	$f(x_0)$
x_1	$f(x_1)$
$\vdots$	$\vdots$
x_n	$f(x_n)$

$$P_n(x) = a_0 + a_1(x-x_0) + a_2(x-x_0)(x-x_1) + a_3(x-x_0)(x-x_1)(x-x_2) + \dots + a_n(x-x_0)(x-x_1)\dots(x-x_{n-1})$$

Here, $a_0 = f[x_0]$

$$a_1 = \frac{f[x_0, x_1] - f[x_0]}{x_1 - x_0} = f[x_0, x_1]$$

$$a_2 = \frac{f[x_0, x_1, x_2] - f[x_0, x_1]}{x_2 - x_1} = f[x_0, x_1, x_2]$$

$$a_n = \frac{f[x_0, x_1, \dots, x_n] - f[x_0, x_1, \dots, x_{n-1}]}{x_n - x_{n-1}} = f[x_0, x_1, \dots, x_n]$$

$$\mu = \frac{f[x_0] - f[x_1]}{x_0 - x_1} = f[x_0, x_1]$$

$$\xi = \frac{f[x_1] - f[x_2]}{x_1 - x_2} = f[x_1, x_2]$$

$$\xi = \frac{f[x_2] - f[x_3]}{x_2 - x_3} = f[x_2, x_3]$$

$$\xi = \frac{f[x_3] - f[x_4]}{x_3 - x_4} = f[x_3, x_4]$$

$$P_n(x) = f[x_0] + f[x_0, x_1](x-x_0) + f[x_0, x_1, x_2](x-x_0)(x-x_1) + \dots + f[x_0, x_1, \dots, x_n](x-x_0)(x-x_1)\dots(x-x_{n-1})$$

$$(1-x)(0-x)((1-x)-x) \cdot 0 + (0-x)((1-x)-x) \cdot \xi + ((1-x)-x) \cdot \mu - \mu = (x) \cdot \xi$$

$$x\xi + \xi x - \mu - \mu =$$

$$2x\xi - \mu =$$

Example:

	x	f(x)
Node=4 Degree=3	$x_0 = -1$	5 $f(x_0)$
	$x_1 = 0$	1 $f(x_1)$
	$x_2 = 1$	3 $f(x_2)$
	$x_3 = 2$	11 $f(x_3)$

New data point $x_4 = 4$ is added between x_3 and x_4

What is $P_n(x)$?

$$\Rightarrow P_3(x) = f[x_0] + f[x_0, x_1](x-x_0) + f[x_0, x_1, x_2](x-x_0)(x-x_1) + f[x_0, x_1, x_2, x_3](x-x_0)(x-x_1)(x-x_2)$$

$$+ f[x_0, x_1, x_2, x_3, x_4](x-x_0)(x-x_1)(x-x_2)(x-x_3)$$

$$x_0 = -1 \quad f[x_0] = 5$$

$$f[x_0, x_1] = \frac{f[x_1] - f[x_0]}{x_1 - x_0} = -4$$

$$x_1 = 0 \quad f[x_1] = 1$$

$$f[x_0, x_1, x_2] = \frac{f[x_1, x_2] - f[x_0, x_1]}{x_2 - x_0} = 3$$

$$f[x_1, x_2] = \frac{f[x_2] - f[x_1]}{x_2 - x_1} = 2$$

$$x_2 = 1 \quad f[x_2] = 3$$

$$f[x_1, x_2, x_3] = \frac{f[x_2, x_3] - f[x_1, x_2]}{x_3 - x_1} = 3$$

$$f[x_2, x_3] = \frac{f[x_3] - f[x_2]}{x_3 - x_2} = 8$$

$$x_3 = 2 \quad f[x_3] = 11$$

$$f[x_2, x_3, x_4] = \frac{f[x_3, x_4] - f[x_2, x_3]}{x_4 - x_2} = \frac{2}{2} = 1$$

$$f[x_3, x_4] = \frac{f[x_4] - f[x_3]}{x_4 - x_3} = \frac{2}{2} = 1$$

$$x_4 = 4 \quad f[x_4] = 20$$

$$P_3(x) = 5 - 4(x - (-1)) + 3(x - (-1))(x - 0) + 0(x - (-1))(x - 0)(x - 1)$$

$$= 5 - 4x - 4 + 3x^2 + 3x$$

$$= 3x^2 - x + 1$$

$$P_4(x) = 5 - 4(x - (-1)) + 3(x - (-1))(x - 0) + 0(x - (-1))(x - 0)(x - 1) - \frac{5}{24}(x - (-1))(x - 0)(x - 1)(x - 2)$$

Cauchy's Theorem

$$|f(x) - P_n(x)| = \frac{f^{(n+1)}(\xi)}{(n+1)!} (x-x_0)(x-x_1) \dots (x-x_n)$$

x_0	$f(x_0)$
x_1	$f(x_1)$
x_n	$f(x_n)$

Error

To find Upper bound error / maximum possible error.

Example: $f(x) = \cos(x)$, interval $\xi \in [-1, 1]$

x	$f(x)$
$-\frac{\pi}{4}$	$\frac{1}{\sqrt{2}}$
0	1
$\frac{\pi}{4}$	$\frac{1}{\sqrt{2}}$

Find the upper bound error.

$$f(x) = \cos x$$

$$f'(x) = -\sin x$$

$$f''(x) = -\cos x$$

$$f'''(x) = \sin x$$

$\Rightarrow$ Node $= n+1 = 3$

Degree $= 2$

$$|f(x) - P_2(x)| = \left| \frac{f^{(3)}(\xi)}{(2+1)!} \left(x + \frac{\pi}{4}\right)(x)(x - \frac{\pi}{4}) \right|$$

$$= \left| \frac{\sin(\xi)}{3!} \left(x + \frac{\pi}{4}\right)(x)(x - \frac{\pi}{4}) \right|$$

$\sin x \rightarrow \left| \frac{\sin(\xi)}{3!} \right| = \left| \frac{\sin(1)}{3!} \right| = 0.140245$

$$w(x) = \left(x + \frac{\pi}{4}\right)\left(x - \frac{\pi}{4}\right)(x)$$

$$w(x) = x^3 - \frac{\pi^2}{16}x$$

$$w'(x) = 3x^2 - \frac{\pi^2}{16} = 0$$

$$\Rightarrow x = \pm \frac{\pi}{4\sqrt{3}}$$

x	$w(x)$
$\frac{\pi}{4\sqrt{3}}$	-0.186474
$-\frac{\pi}{4\sqrt{3}}$	0.186474
-1	-0.383150
1	0.383150

To get max value of a function

$$\frac{dy}{dx} = 0$$

$$\therefore w'(x) = 0, x \Rightarrow \text{maximum}$$

Upper bound error = $|0.1402545 \times 0.383150| = 0.0537349$

→ Hermite Interpolation

	x	$f(x)$	$f'(x)$
Node = $n+1$	x_0	$f(x_0)$	$f'(x_0)$
	x_1	$f(x_1)$	$f'(x_1)$
	$\vdots$	$\vdots$	$\vdots$
degree = $(n+1)-1$ $= n$	x_n	$f(x_n)$	$f'(x_n)$

$(x_0, f(x_0)), (x_1, f(x_1)), \dots, (x_n, f(x_n))$ total $(n+1)$ nodes/conditions

$(x_0, f'(x_0)), (x_1, f'(x_1)), \dots, (x_n, f'(x_n))$ total $(n+1)$ nodes/conditions

Total = $2n+2$ nodes/condition

∴ Hermite polynomial Degree = $(2n+2)-1$

Hermite interpolation, $\left| \left(\frac{x}{p} - x \right) (x) \left(\frac{x}{p} + x \right) \frac{(x)^{n+2}}{(1+x)^{n+2}} \right| = \left| (x)^{n+2} - (x)^{n+2} \right|$

$P_{2n+1}(x) = \sum_{k=0}^n h_k(x) f(x_k) + \sum_{k=0}^n \hat{h}_k(x) f'(x_k)$

$h_k(x) = [1 - 2(x-x_k) \{L'_k(x_k)\}] \{L_k(x)\}^2$

$\hat{h}_k(x) = (x-x_k) \{L_k(x)\}^2$

$0 = \frac{y_b}{x_b}$

$(x)w$	x
-0.186474	$\frac{x}{2p}$
0.186474	$\frac{x}{2p}$
-0.383150	-1
0.383150	1

$2p \sin \theta = \left| \frac{(1) \sin \theta}{18} \right| = \left| \frac{(x) \sin \theta}{18} \right| + x \sin \theta$

$(x) \left(\frac{x}{p} - x \right) \left(\frac{x}{p} + x \right) = (x)w$

$x \frac{x}{2p} = x(x)w$

$0 = x \frac{x}{2p} - x(x)w = (x)'w$

$\frac{x}{2p} \pm x = x$

Example: Given function $f(x) = \sin x$ and nodes $\{0, \frac{\pi}{2}\}$, find out the hermite polynomial.

x	$f(x)$	$f'(x) = \cos x$
$x_0 = 0$	$0 = f(x_0)$	$1 = f'(x_0)$
$x_1 = \frac{\pi}{2}$	$1 = f(x_1)$	$0 = f'(x_1)$

Nodes = $2n+2$
Degree = $2n+1$
 $n=1$

$$P_{2n+1} = P_{2n+1} = P_3(x) = h_0(x)f(x_0) + \hat{h}_0(x)f'(x_0) + h_1(x)f(x_1) + \hat{h}_1(x)f'(x_1)$$

$$(x) \{ (x) \} \hat{h} = \hat{h}_0(x)f'(x_0) + h_1(x)f'(x_1)$$

$$\hat{h}_0(x) = (x-x_1) \{ l_1(x) \}^2$$

$$= (x-x_0) \{ l_0(x) \}^2$$

$$l_0(x) = \frac{x-x_1}{x_0-x_1} = \frac{x-\frac{\pi}{2}}{-\frac{\pi}{2}}$$

$$= (x-0) \left\{ \frac{x-\frac{\pi}{2}}{-\frac{\pi}{2}} \right\}^2$$

$$= x \left\{ \frac{x-\frac{\pi}{2}}{-\frac{\pi}{2}} \right\}^2$$

$$h_1(x) = \left[1 - 2(x-x_0) \{ l'_0(x_0) \} \right] \{ l_0(x) \}$$

$$= \left[1 - 2(x-x_1) \{ l'_1(x_1) \} \right] \{ l_1(x) \}^2$$

$$= \left[1 - 2(x-\frac{\pi}{2}) \left(\frac{2}{\pi} \right) \right] \left(\frac{2x}{\pi} \right)^2$$

$$P_3(x) = x \left\{ \frac{x-\frac{\pi}{2}}{-\frac{\pi}{2}} \right\}^2 (1) + \left[1 - 2(x-\frac{\pi}{2}) \left(\frac{2}{\pi} \right) \right] \left(\frac{2x}{\pi} \right)^2 (1)$$

$$(1-x) \frac{1}{2} = (x) \frac{1}{2}$$

$$2x-1 = (x) \frac{1}{2}$$

$$x \frac{1}{2} - 1 = (x) \frac{1}{2}$$

$$x \frac{1}{2} - 1 = (x) \frac{1}{2}$$

$$l_1(x) = \frac{x-x_0}{x_1-x_0} = \frac{x-0}{\frac{\pi}{2}-0} = \frac{2x}{\pi}$$

$$l'_1(x) = \frac{2}{\pi}$$

$$l'_1(x_1) = l'_1\left(\frac{\pi}{2}\right) = \frac{2}{\pi}$$

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Example:

	x	f(x)	f'(x)
Node = m+1	-1	1	2
i=3	0	0	2
n=2	1	1	0

Total node = $2n+2 = 6$
 Degree = $2n+1$

Using Hermite Polynomial,

$$P_{2n+1}(x) = \hat{h}_0(x)f(x_0) + \hat{h}_1(x)f'(x_0) + \hat{h}_2(x)f(x_1) + \hat{h}_3(x)f'(x_1)$$

$$+ \hat{h}_2(x)f(x_2) + \hat{h}_2(x)f'(x_2)$$

$$\Rightarrow P_5(x) = h_0(x)(1) + \hat{h}_0(x)(2) + \hat{h}_1(x)(2) + h_2(x)(1)$$

$$\frac{x}{2} - x = \frac{1x - x}{1x - 0x} = h_0(x) + 2\hat{h}_0(x) + 2\hat{h}_1(x) + h_2(x)$$

$$h_0(x) = x - [1 - 2(x-x_0)\{l'_k(x_k)\}] \{l_k(x)\}^2$$

$$= [1 - 2(x-x_0)\{l'_0(x_0)\}] \{l_0(x)\}^2$$

$$= [1 - 2(x+1)(-\frac{3}{2})] \{\frac{1}{2}(x^2-x)\}^2$$

$$\hat{h}_0(x) = (x-x_k) \{l_k(x)\}^2$$

$$= (x-x_0) \{l_0(x)\}^2$$

$$= (x+1) \{\frac{1}{2}(x^2-x)\}^2$$

$$h_1(x) = [1 - 2(x-x_k)\{l'_k(x_k)\}] \{l_k(x)\}^2$$

$$= [1 - 2(x-x_1)\{l'_1(x_1)\}] \{l_1(x)\}^2$$

$$\hat{h}_1(x) = (x-x_k) \{l_k(x)\}^2$$

$$= (x-x_1) \{l_1(x)\}^2$$

$$= (x-0) (1-2x)^2$$

$$= x (1-2x)^2$$

$$l_0(x) = \frac{x-x_1}{x_0-x_1} \cdot \frac{x-x_2}{x_0-x_2}$$

$$= \frac{x-0}{-1-0} \cdot \frac{x-1}{0-1-1}$$

$$= \frac{x}{-1} \cdot \frac{x-1}{-2} = \frac{x(x-1)}{2}$$

$$l'_0(x) = \frac{1}{2}(2x-1)$$

$$l'_0(x_0) = \frac{1}{2}(-2-1) = -\frac{3}{2}$$

$$l_1(x) = \frac{x-x_0}{x_1-x_0} \cdot \frac{x-x_2}{x_1-x_2}$$

$$= \frac{x+1}{1} \cdot \frac{x-1}{-1}$$

$$= -(x^2-1)$$

$$= 1-x^2$$

$$l'_1(x) = 1-2x$$

$$l'_1(x_1) = 1-2 \cdot 0 = 1$$

$$h_2(x) = [1 - 2(x-x_2)\{l'_2(x_2)\}] \{l_2(x)\}^2$$

$$= [1 - 2(x-1)(\frac{3}{2})] \{\frac{1}{2}(x^2+x)\}^2$$

$$l_2(x) = \frac{x-x_0}{x_2-x_0} \cdot \frac{x-x_1}{x_2-x_1}$$

$$= \frac{x+1}{1+1} \cdot \frac{x-0}{1-0}$$

$$l'_2(x) = \frac{1}{2}(2x+1)$$

$$l'_2(1) = \frac{3}{2}$$

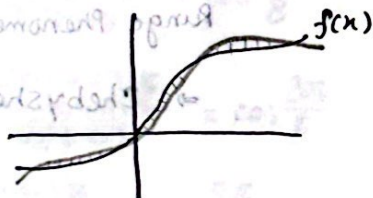
$$\therefore P_5(x) = [1 - 2(x+1)(-\frac{3}{2})] \{\frac{1}{2}(x^2-x)\}^2$$

$$+ 2(x+1)\{\frac{1}{2}(x^2-x)\}^2 + 2x(1-2x)\{\frac{1}{2}(x^2-x)\}^2$$

$$+ [1 - 2(x-1)(\frac{3}{2})] \{\frac{1}{2}(x^2+x)\}^2$$

Advantage:

⇒ Convergence:



- is how closely the polynomial is covering the function

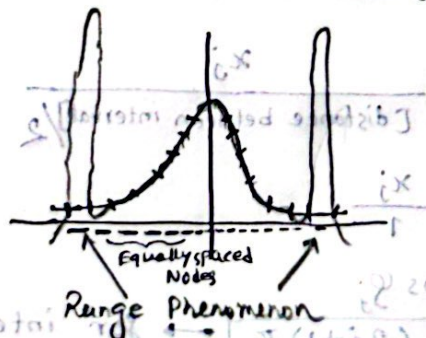
⇒ Runge Phenomenon:

According to Weierstrass Method

the $n \uparrow$ error $\downarrow$

but its not always true.

(Not true for Runge function)



Runge phenomenon Depends on 2 things

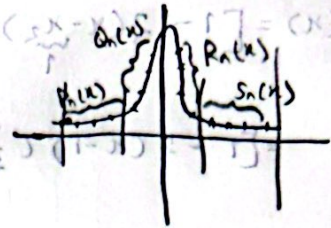
① Depends on the function (Symmetric function)

② Depends on the Node (Equally spaced)

Solution:

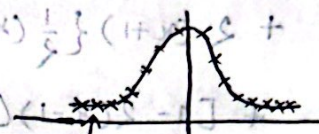
① Piece wise interpolation:

- Take smaller intervals and find the interpolating polynomial
- Add/Merge them up.



② No equidistant Node:

- Chebyshev's Node



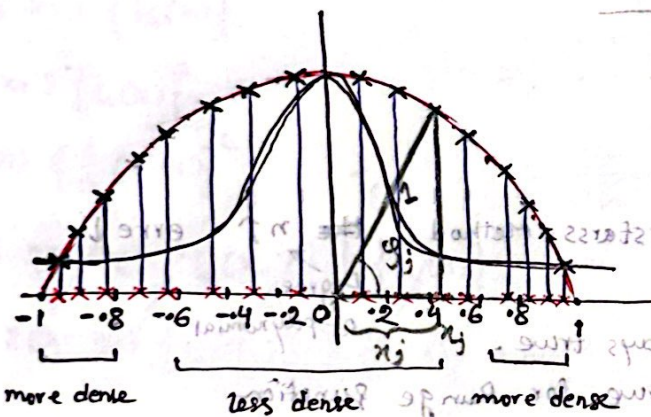
interval taking closely in Runge region

↔ Chebyshev's Node:

- * Takes more node at the endpoint rather than taking equidistance Node

How do we find these perfect nodes such that we get least amount of error and reduce the effect of Runge Phenomenon?

→ Chebyshev's Node



Runge Phenomenon

$j = j^{th}$ Node

$$y_j = \frac{(2j+1)\pi}{2(n+1)}$$

↑ Degree

$$\cos y_j = \frac{x_j}{[\text{distance between interval}]/2}$$

$$\cos y_j = \frac{x_j}{1}$$

$$\Rightarrow x_j = \cos y_j$$

$$x_j = \cos \frac{(2j+1)\pi}{2(n+1)}$$

→ for interval $[-1, 1]$ only and center at 0.

For other intervals,

$$x_j = \cos \frac{2(j+1)\pi}{2(n+1)}$$

$$x_j = \frac{(2j+1)\pi}{2(n+1)} \times \underbrace{\text{radius}}_{\substack{\text{interval dist} \\ 2}} + \text{center}$$

Example: $f(x) = \frac{1}{1+25x^2}$, $[-1,1]$, $n=3$. Find Chebyshev's node

$\Rightarrow n=3 \rightarrow \text{Node} = 4 \therefore j = 0, 1, 2, 3$
 $x_0 \quad x_1 \quad x_2 \quad x_3$

$$x_j = \cos \frac{(2j+1)\pi}{2(n+1)}$$

$$x_0 = \cos \frac{(2 \times 0 + 1)\pi}{2(3+1)} = \cos \frac{\pi}{8}$$

$$x_1 = \cos \frac{(2 \times 1 + 1)\pi}{2(3+1)} = \cos \frac{3\pi}{8}$$

$$x_2 = \cos \frac{(2 \times 2 + 1)\pi}{2(3+1)} = \cos \frac{5\pi}{8}$$

$$x_3 = \cos \frac{(2 \times 3 + 1)\pi}{2(3+1)} = \cos \frac{7\pi}{8}$$

79 $\rightarrow$ Using the nodes find interpolating polynomial using Lagrange / Vandermonde / Newton / Hermite interpolation.